Mix proportioning of concrete (IS 10262, HSC and ACI methods)

Notes from the lecture slides "Module 4 – Mix proportioning of concrete (Concrete Mix Design), HSC, ACI Mix design"

In this module

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1 What mix proportioning is for

Mix proportioning (mix design) is choosing the right amounts of cement, mineral admixtures, water, fine and coarse aggregate and chemical admixtures so that the concrete does its job at the lowest cost. Concrete can be tailored to almost any structure and shape: sewer pipes, the Burj Khalifa (2009), the Bahá'í (Lotus) Temple in New Delhi (1986), the Tenerife Auditorium (2003).

1.1 Requirements of good concrete

1. Meet the **strength** required for the application.
2. Meet **durability** requirements for the environment it will serve in.
3. Be **mixed, transported and compacted** as efficiently as possible.
4. Be as **economical and sustainable** as possible.

1.2 Requirements in the fresh state

- Easy to **mix and transport**
- **Uniform** within a batch and between batches
- Able to **fill the formwork completely**
- Can be **placed and fully compacted** without a lot of energy
- **No segregation** during placing and compaction
- Can be **finished properly**

1.3 Information needed before designing a mix (IS 10262:2019)

- Grade designation (e.g. M40)
- Type and grade of cement
- Maximum nominal size of aggregate
- Minimum cement / cementitious content and **maximum cement content**
- Maximum water-cementitious ratio
- **Workability** needed at placing (slump)
- Transportation time and **method of placing** (chute, pump…)
- **Degree of site control** (good or fair) or an established standard deviation
- Type of coarse aggregate (angular, sub-angular, gravel with crushed particles, rounded gravel, manufactured)
- Type of fine aggregate (natural sand, crushed stone/gravel sand, manufactured sand, mixed sand)
- Whether **chemical and mineral admixtures** are to be used, their type and amount
- Any other requirement, e.g. **early-age strength**

2 Exposure conditions and durability limits (IS 456:2000)

Table 3: Environmental exposure conditions

Environment	Exposure conditions
Mild	Concrete surfaces protected against weather or aggressive conditions, except those in a coastal area
Moderate	Sheltered from severe rain or freezing whilst wet; exposed to condensation and rain; continuously under water; in contact with or buried under non-aggressive soil/groundwater; sheltered from saturated salt air in a coastal area
Severe	Exposed to severe rain, alternate wetting and drying, or occasional freezing whilst wet or severe condensation; completely immersed in sea water; exposed to a coastal environment
Very severe	Exposed to sea water spray, corrosive fumes, or severe freezing whilst wet; in contact with or buried under aggressive sub-soil/groundwater
Extreme	Surface of members in the tidal zone; members in direct contact with liquid or solid aggressive chemicals

Table 5: Minimum cement content, maximum w/c and minimum grade (20 mm aggregate)

Exposure	Plain: min cement (kg/m³)	Plain: max w/c	Plain: min grade	RCC: min cement (kg/m³)	RCC: max w/c	RCC: min grade
Mild	220	0.60	–	300	0.55	M20
Moderate	240	0.60	M15	300	0.50	M25
Severe	250	0.50	M20	320	0.45	M30
Very severe	260	0.45	M20	340	0.45	M35
Extreme	280	0.40	M25	360	0.40	M40

The cement content is irrespective of the cement grade and includes mineral additions (fly ash, GGBS) within the limits of IS 1489 and IS 455. IS 456 also caps OPC content at **450 kg/m³** (excluding fly ash and slag).

3 Characteristic strength and target mean strength

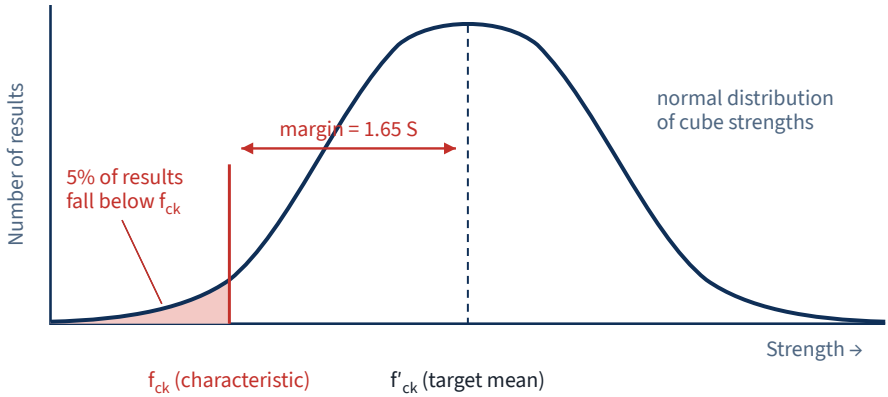
3.1 Why we design for a higher strength

Cube strengths from the same mix **vary** and follow roughly a **normal distribution** around their mean. We cannot guarantee that *no* cube will fall below a value, so design uses a **"safe" strength**: one that only a small, acceptable fraction of results fall below. This is the **characteristic strength**:

$$\sigma_{char} = \sigma_m - k\,s$$

where σ_m = mean strength, s = standard deviation and k = a probability factor.

- k = 0 → 50% of results fall below (the mean itself).
- **k = 1.64 (≈ 1.65) → 5% fall below** (1 in 20). This is the value used in IS codes.



The mix is designed for the target mean f'_{ck} so that only 5% of results fall below the specified f_{ck}.

k	% of results within $f_m \pm ks$	% below $f_m - ks$ (risk)
1.00	68.2	15.9 (1 in 6)
1.64	90.0	**5.0 (1 in 20)**
1.96	95.0	2.5 (1 in 40)
2.33	98.0	1.0 (1 in 100)
3.00	99.7	0.15 (1 in 700)

3.2 Target mean strength (IS 10262:2019)

Target strength: take the higher of

$$f'_{ck} = f_{ck} + 1.65\,S \quad \text{or} \quad f'_{ck} = f_{ck} + X$$

f'_{ck} = target mean compressive strength at 28 days; f_{ck} = characteristic strength (the grade, e.g. 40 for M40); S = standard deviation; X = factor based on grade.

Grade	X (Table 1)	Assumed S, good control (Table 2)
M10, M15	5.0	3.5
M20, M25	5.5	4.0
M30 to M60	6.5	5.0
M65 and above	8.0	6.0

- The tabulated S is for **good** site control (proper storage of cement, weigh batching, controlled water, regular checks of materials, grading and moisture, workability and strength). For **fair** control, **increase S by 1 N/mm²**.
- **Using site data:** S should be based on **at least 30** test results. Recalculate it for each significant change in materials, and keep it **up to date**, checking periodically and after every change in mix proportions.

4 IS 10262:2019 method for ordinary and standard grades

The steps:

1. Target strength
2. Water–cement ratio
3. Air content
4. Water content (and admixture content)
5. Cement / cementitious content
6. Coarse aggregate proportion
7. Combining coarse aggregate fractions
8. Fine and coarse aggregate contents
9. Trial mixes

4.1 Step 1: Water–cement ratio

The strength–w/c relationship depends on the **type of cement, SCMs, aggregate size, grading, surface texture and shape**. Ideally it should be **established for the actual materials**. Without such data, use **Fig. 1 of IS 10262**, which has three curves for the 28-day strength of the cement:

- **Curve 1**: cement strength 33 to < 43 N/mm² (OPC 33)
- **Curve 2**: 43 to < 53 N/mm² (OPC 43, and PPC/PSC)
- **Curve 3**: 53 N/mm² and above (OPC 53)

Read the w/c for the target strength from the right curve, then **check it against the maximum w/c in IS 456 Table 5** and adopt the lower value.

Approximate readings from the chart:

Free w/c	0.36	0.38	0.42
Curve 3 (OPC 53)	~54	~52	~46
Curve 2 (OPC 43, PPC)	~47	~44	~39
Curve 1 (OPC 33)	~41	~39	~34

4.2 Step 2: Air content

Approximate **entrapped air** (non-air-entrained concrete), IS 10262 Table 3:

Nominal max. size of aggregate	Entrapped air (% of volume)
10 mm	1.5
20 mm	**1.0**
40 mm	0.8

Actual site values can be used if available for a similar mix (at least 5 results).

4.3 Step 3: Water content and admixture

Factors affecting water demand: aggregate size, shape and texture; workability; w/c ratio; cement/SCM type and content; environmental conditions.

- **Less water** is needed with **larger aggregate**, **rounded aggregate** and **water-reducing admixtures**.
- **More water** is needed with **higher temperature**, **higher cement content**, **more angular aggregate**, and a **lower proportion of CA to FA**.

Water content per m³ (Table 4), for **angular** coarse aggregate and **50 mm slump**:

Nominal max. size	Water (kg/m³)
10 mm	208
20 mm	**186**
40 mm	165

Adjustments:

- **Aggregate shape:** reduce water by **10 kg** for sub-angular, **15 kg** for gravel with some crushed particles, **20 kg** for rounded gravel.
- **Slump:** increase (or decrease) water by about **3% for every 25 mm** increase (or decrease) in slump.
- **Admixtures:** water-reducing admixtures cut water by **5–10%**; superplasticizers by **20–30%**.

- High dosages of mineral admixtures may **increase** water or admixture demand.
- **Ready-mix over long distances:** design for a **higher initial slump**, depending on transport distance and slump loss with time.
- Always confirm with **trial batches**, because aggregates from different sources behave differently.

4.4 Step 4: Cement (cementitious) content

$$\text{Cement content} = \frac{\text{water content}}{\text{w/c ratio}}$$

- If **fly ash, GGBS, silica fume** etc. are used, check the part replacement (especially if fly ash is more than 20%); the cementitious content is usually increased and the **w/cm recalculated**.
- **Check against IS 456**: not less than the **minimum**, and OPC not more than the **maximum** (450 kg/m^3).
- The % of fly ash or GGBS is decided by project needs and the quality of these materials.

4.5 Step 5: Coarse aggregate proportion

Table 5 of IS 10262: volume of coarse aggregate per unit volume of **total aggregate**, for **w/c = 0.50**:

Nominal max. size	Zone IV	Zone III	Zone II	Zone I
10 mm	0.54	0.52	0.50	0.48
20 mm	0.66	0.64	**0.62**	0.60
40 mm	0.73	0.72	0.71	0.69

(Volumes are for SSD aggregate and **crushed angular** CA. Zone IV sand is not recommended for reinforced concrete without trials.)

Adjustments:

- For other w/c ratios: **increase the CA proportion by 0.01 for every 0.05 decrease in w/c** (and decrease it by 0.01 for every 0.05 increase). (The slide

prints "decrease" in both lines; the second should read "increase in w/c".)

- For equal workability the CA volume depends on its **maximum size** and the **grading zone of the sand**.
- Crushed angular CA needs **more fine aggregate** (more surface area to coat).
- For **pumpable concrete** or congested reinforcement, the CA content may be **reduced by up to 10%**. Check that the slump, w/c and strength still meet IS 456 and the project specification.

4.6 Step 6: Combining coarse aggregate fractions

CA must conform to **IS 383**. Single-size fractions are blended to give the overall grading in **Table 7 of IS 383**. For example, for 20 mm graded aggregate:

IS sieve (mm)	Fraction I (20–10 mm), % passing	Fraction II (10–4.75 mm), % passing	60% of I	40% of II	Combined
20	100	100	60	40	100
10	0	71.20	0	28.5	28.5
4.75	–	9.40	–	3.7	3.7
2.36	–	0	–	–	–

The combined grading conforms to Table 7 of IS 383.

4.7 Step 7: Fine and coarse aggregate contents

All quantities except aggregate are now known. For **1 m³** of concrete:

	Volume
a	Total volume = 1 m^3
b	Volume of entrapped air (from Step 2)
c	Volume of cement = mass of cement ÷ (SG of cement × 1000)
d	Volume of water = mass of water ÷ 1000
e	Volume of chemical admixture = mass ÷ (SG × 1000)
f	**Volume of all aggregate = (a − b) − (c + d + e)**

Split f into CA and FA using the CA proportion from Step 5, then:

$$\text{Mass of CA} = f \times (\text{CA proportion}) \times SG_{CA} \times 1000$$

$$\text{Mass of FA} = f \times (\text{FA proportion}) \times SG_{FA} \times 1000$$

(The slide writes SG_{CA} in the FA formula by mistake; use the **specific gravity of the fine aggregate** for FA.)

4.8 Step 8: Adjusting for aggregate moisture

The design gives aggregate masses in the **SSD** state. If the aggregate on site is **dry**, it will absorb water from the mix, so:

$$\text{Dry mass} = \frac{\text{SSD mass}}{1 + (\text{water absorption \%} \div 100)}$$

- Extra water for CA = SSD mass − dry mass of CA; extra water for FA likewise.
- **Total water to add = design water + extra water for CA + extra water for FA.**
- If the aggregate is **wet** (free surface moisture), the free water is **subtracted** from the mixing water and the aggregate mass increased by the same amount.

4.9 Step 9: Trial mixes

Make a trial batch with these proportions, check **workability, cohesion and strength**, and adjust (water/admixture for slump; w/c for strength) until the requirements are met.

5 Worked example: M40 (from the class)

Data: M40; **PPC** (IS 1489 Part 1); 20 mm max size; **Zone II** sand; **severe** exposure, reinforced; slump **75 mm**; placing by chute; **good** site control; crushed angular aggregate; max cement 450 kg/m^3; superplasticizer (normal).

Test data: SG of cement (PPC) 2.88; CA (SSD) 2.74; FA (SSD) 2.65; chemical admixture 1.145. Water absorption: CA 0.5%, FA 1.0%.

1. **Target strength**
   - $f'_{ck} = 40 + 1.65 \times 5 =$ **48.25 MPa**

- f'_{ck} = 40 + 6.5 = 46.5 MPa
- Take the higher: **48.25 MPa**.

2. **Air content** (20 mm): **1.0%**.
3. **w/c ratio:** from Curve 2 (PPC) for 48.25 MPa → **0.36**. IS 456 maximum for severe RCC = 0.45 > 0.36, so **adopt 0.36**.
4. **Water content:** 186 kg (20 mm, 50 mm slump). For 75 mm (+25 mm): 186 × 1.03 = 191.58 kg. Superplasticizer reduces water by 23%: 191.58 × 0.77 = 147.5 ≈ **148 kg/m³**.
5. **Cement:** 148 ÷ 0.36 = **411.11 kg/m³**. Minimum for severe RCC = 320 kg/m³ and maximum 450, so OK.
6. **CA proportion:** Table 5, 20 mm, Zone II → 0.62 at w/c 0.5. w/c is lower by 0.14, so add 0.01 × (0.14 ÷ 0.05) = 0.028 → **0.648**. FA proportion = 1 − 0.648 = **0.352**.
7. **Volumes per m³:**
   - Air = 0.01 m³
   - Cement = 411.11 ÷ (2.88 × 1000) = 0.143 m³
   - Water = 148 ÷ 1000 = 0.148 m³
   - Admixture (1% of cement = 4.11 kg) = 4.11 ÷ (1.145 × 1000) = 0.0036 m³
   - All aggregate = (1 − 0.01) − (0.143 + 0.148 + 0.0036) = **0.695 m³**
8. **Aggregate masses (SSD):**
   - CA = 0.695 × 0.648 × 2.74 × 1000 = **1234 kg**
   - FA = 0.695 × 0.352 × 2.65 × 1000 = **648 kg**
9. **Adjust for dry aggregate:**
   - FA (dry) = 648 ÷ 1.01 = 641.6 ≈ **642 kg**; extra water = 648 − 642 = 6 kg
   - CA (dry) = 1234 ÷ 1.005 = **1228 kg**; extra water = 1234 − 1228 = 6 kg
   - Water to add = 148 + 6 + 6 = **160 kg**

Material (per m^3)	SSD aggregates	Dry aggregates (as batched)
Cement (PPC)	411.11 kg	411.11 kg
Water	148 kg	**160 kg**
Fine aggregate	648 kg	642 kg
Coarse aggregate	1234 kg	1228 kg
Chemical admixture	4.11 kg	4.11 kg
Free w/c	0.36	0.36

6 Mix design for high-strength concrete (M65 and above)

6.1 Materials for HSC

- **Cementitious materials:** specially selected; **fly ash, silica fume, GGBS or metakaolin** are widely used as cementitious and pozzolanic ingredients (they also have a filler effect).
- **Coarse aggregate:** strong, sound, free of fissures or weak planes, clean, no surface coatings. Crushed aggregate with **impact/crushing value ≤ 22%** and **combined flakiness and elongation index ≤ 30%**.
- **Fine aggregate: coarser sand (Zone I or II)** is preferred, since the high cementitious content already supplies plenty of fines.
- **Chemical admixtures:** a **low w/c** is only achievable with **high-range water reducers (HRWR)**, which reduce water by about **30%** (PCE based).

6.2 Procedure (similar to ordinary concrete)

- **Silica fume** is generally advantageous for **M50 and above**, and for high-performance concrete with special needs such as **abrasion resistance**.
- **Target strength:** same formulas (S = 6.0, X = 8.0 for M65+).
- **Maximum aggregate size: 20 mm**; for **M80 and above, 10–12.5 mm** may be preferable.
- **Air content (Table 6):** 10 mm → 1.0%; 12.5 mm → 0.8%; **20 mm → 0.5%**.

Water content for HSC without admixtures, 50 mm slump (Table 7):

Nominal max. size	Max. water (kg/m^3)
10 mm	200
12.5 mm	195
20 mm	**186**

Adjust by ~3% per 25 mm slump; higher mineral admixture dosages may need more water or admixture.

Recommended w/cm for HSC with HRWR (Table 8) (for 53 MPa cement):

Target strength (N/mm^2)	10 mm	12.5 mm	20 mm
70	0.36	0.35	0.33
75	0.34	0.33	0.31
80	0.32	0.31	0.29
85	0.30	0.29	0.27
90	0.28	0.27	0.26
100	0.26	0.25	0.24

These assume **silica fume** is used. For other cementitious materials, **increase the cementitious content** and recalculate w/cm.

Cementitious content = water ÷ w/cm, and it must satisfy IS 456 limits. If the OPC content (excluding mineral admixtures) must exceed the IS 456 maximum, special consideration must be given to the higher risk of **drying shrinkage cracking**, **early thermal cracking** and **alkali–silica reaction**.

Recommended mineral admixture dosages for HSC (Table 9), % by mass of total cementitious material:

Mineral admixture	Dosage
Fly ash	15–30%
GGBS	25–50%
Metakaolin	5–15%
Silica fume	5–10%

(Note from class: too much silica fume makes the mix harsh, so its dosage is kept low.)

Coarse aggregate proportion (Table 10), per unit volume of total aggregate at **w/cm = 0.30**:

Nominal max. size	Zone III	Zone II	Zone I
10 mm	0.56	0.54	0.52
12.5 mm	0.58	0.56	0.54
20 mm	0.68	**0.66**	0.64

For **pumpable** concrete the CA may be reduced by **up to 5%**. Fine and coarse aggregate contents are then found exactly as for ordinary concrete.

7 Worked example: M70 high-strength concrete (from the class)

Data: M70; OPC 53 (IS 269); silica fume (IS 15388); 20 mm; severe exposure, reinforced; slump **120 mm**; **pumping**; good supervision; crushed angular aggregate; max OPC 450 kg/m^3; **PCE superplasticizer**. Mineral admixtures: **fly ash 15%** and **silica fume 5%** of total cementitious material.

1. **Target strength:** 70 + 1.65 × 6 = **79.9 MPa**; 70 + 8 = 78 MPa → take **79.9 MPa**.
2. **Air content** (20 mm): **0.5%**.
3. **w/cm:** Table 8, 20 mm: 75 → 0.31, 80 → 0.29; for 79.9 → **0.29**.
4. **Water:** 186 kg for 50 mm slump. For 120 mm: (120 − 50) ÷ 25 × 3% = 8.4% → 186 × 1.084 = 201.6 ≈ 202 kg. PCE reduces water by 30%: 202 × 0.70 = 141.4 ≈ **141**

kg/m³.

5. **Cementitious content:** 141 ÷ 0.29 = 486 kg/m³. Because fly ash is also used (not just silica fume), increase by 10%: 486 × 1.1 ≈ **535 kg/m³**.
   - Fly ash (15%) = 0.15 × 535 = **80.25 kg**
   - Silica fume (5%) = 0.05 × 535 = **26.75 kg**
   - OPC = 535 − 80.25 − 26.75 = **428 kg** (< 450 maximum, > 320 minimum: OK)
   - Revised w/cm = 141 ÷ 535 = **0.264**
6. **CA proportion:** Table 10, 20 mm, Zone II → 0.66 at w/cm 0.30. w/cm is lower by 0.036 → add 0.01 × (0.036 ÷ 0.05) ≈ 0.007 → **0.667**; FA = **0.333**.

The class slides stop after listing the specific gravities (cement 3.15, silica fume 2.2, fly ash 2.2, admixture 1.08). The remaining steps are completed below, **assuming** SG of CA = 2.74 and FA = 2.65 (as in the M40 example) and a PCE dosage of 1.5% by mass of cementitious material (the slide's note is hard to read). Use your teacher's values if they differ.

7. **Volumes per m³:**
   - Air = 0.005 m³
   - OPC = 428 ÷ 3150 = 0.1359 m³
   - Silica fume = 26.75 ÷ 2200 = 0.0122 m³
   - Fly ash = 80.25 ÷ 2200 = 0.0365 m³
   - Water = 0.141 m³
   - Admixture = 8.03 kg ÷ 1080 = 0.0074 m³
   - All aggregate = (1 − 0.005) − 0.3330 = **0.662 m³**
8. **Masses (SSD):**
   - CA = 0.662 × 0.667 × 2.74 × 1000 ≈ **1210 kg**
   - FA = 0.662 × 0.333 × 2.65 × 1000 ≈ **584 kg**

Material (per m^3)	Quantity
OPC 53	428 kg
Fly ash	80.25 kg
Silica fume	26.75 kg
Water	141 kg
Fine aggregate (SSD)	≈ 584 kg
Coarse aggregate (SSD)	≈ 1210 kg
PCE superplasticizer (assumed 1.5%)	≈ 8 kg
w/cm	0.264

8 Exercise from the slides: M35 (solved)

Data: M35; **OPC 53**; 20 mm; Zone II; **very severe** exposure, reinforced; slump **70 mm**; chute; **fair** site control; crushed angular aggregate; max cement 450 kg/m^3. No chemical admixture is specified.

Assumed (not given on the slide): SG cement 3.15, CA 2.74, FA 2.65; absorption CA 0.5%, FA 1.0%; aggregates dry.

1. **Target strength:** S = 5 + 1 (fair control) = 6 N/mm^2. f'_{ck} = 35 + 1.65 × 6 = **44.9 MPa**; 35 + 6.5 = 41.5 → take **44.9 MPa**.
2. **Air:** 1.0%.
3. **w/c:** Curve 3 (OPC 53) at 44.9 MPa → about **0.43**. IS 456 max for very severe RCC = 0.45 → adopt **0.43**.
4. **Water:** 186 × (1 + 0.03 × 20 ÷ 25) = 186 × 1.024 = **190.5 kg**.
5. **Cement:** 190.5 ÷ 0.43 = **443 kg/m^3** (≥ 340 minimum, ≤ 450 maximum: OK, though close to the limit; a plasticizer would lower it).
6. **CA proportion:** 0.62 + 0.01 × (0.07 ÷ 0.05) = **0.634**; FA = 0.366.
7. **Aggregate volume:** 0.99 − (443 ÷ 3150 + 0.1905) = 0.99 − (0.1406 + 0.1905) = **0.659 m^3**.

8. **Masses (SSD):** CA = 0.659 × 0.634 × 2.74 × 1000 ≈ **1145 kg**; FA = 0.659 × 0.366 × 2.65 × 1000 ≈ **639 kg**.
9. **Dry aggregate:** CA = 1145 ÷ 1.005 ≈ 1139 kg; FA = 639 ÷ 1.01 ≈ 633 kg; water = 190.5 + 6 + 6 ≈ **202 kg**.

Proportions (dry aggregate): cement 443 : water 202 : FA 633 : CA 1139 kg/m^3, free w/c 0.43.

9 ACI 211.1-91 method

9.1 Specifications needed

- Maximum w/c (or w/cm) ratio
- Minimum cement content
- Air content
- Slump
- Maximum aggregate size
- Strength
- Other requirements: strength over-design, admixtures, special cements, other cementitious materials or aggregates

9.2 Required average strength (ACI 318)

Cylinder strengths are used in ACI (cylinder strength ≈ 0.8–0.85 × cube strength).

With a known standard deviation s:

- For $f'_c \le 35$ MPa: f'_{cr} = the larger of $\mathbf{f'_c + 1.34\,s}$ and $\mathbf{f'_c + 2.33\,s - 3.5}$
- For $f'_c > 35$ MPa: the larger of $\mathbf{f'_c + 1.34\,s}$ and $\mathbf{0.90\,f'_c + 2.33\,s}$

If fewer than 30 tests are available, multiply s by:

Number of tests	Factor for s
15	1.16
20	1.08
25	1.03
30 or more	1.00

(If there are no data at all, ACI 318 uses $f'_{cr} = f'_c + 7$ MPa for $f'_c < 21$ MPa, $f'_c + 8.3$ MPa for 21–35 MPa, and $1.10\, f'_c + 5$ MPa above 35 MPa.)

9.3 Step 1: Slump

If not specified, choose from ACI Table A1.6.3.1 (for vibrated concrete):

Type of construction	Max. slump (mm)	Min. slump (mm)
Reinforced foundation walls and footings	75	25
Plain footings, caissons, substructure walls	75	25
Beams and reinforced walls	100	25
Building columns	100	25
Pavements and slabs	75	25
Mass concrete	50	25

Slump may be increased by 25 mm for compaction methods other than vibration. Take care with chemical admixtures to avoid segregation or bleeding.

9.4 Step 2: Maximum aggregate size

- Use the **largest size that is economically available** and consistent with the dimensions of the structure.
- In areas **congested** with reinforcement, ducts or conduits, choose a size that lets the concrete be placed without segregation, pockets or voids.
- For **high-strength concrete**, **smaller** maximum sizes give higher strength at a given w/c.

9.5 Step 3: Mixing water and air content

ACI Table A1.6.3.3, water in kg/m^3:

Slump (mm)	9.5	12.5	19	25	37.5	50	75	150 mm
Non-air-entrained								
25–50	207	199	190	179	166	154	130	113
75–100	228	216	**205**	193	**181**	169	145	124
150–175	243	228	216	202	190	178	160	–
Entrapped air (%)	3	2.5	2	1.5	1	0.5	0.3	0.2
Air-entrained								
25–50	181	175	168	160	150	142	122	107
75–100	202	193	184	175	165	157	133	119
150–175	216	205	197	184	174	166	154	–
Total air, mild (%)	4.5	4.0	3.5	3.0	2.5	2.0	1.5	1.0
Total air, moderate (%)	6.0	5.5	5.0	4.5	4.5	4.0	3.5	3.0
Total air, extreme (%)	7.5	7.0	6.0	6.0	5.5	5.0	4.5	4.0

Corrections (table notes): values are for well-shaped angular aggregate; **rounded CA needs about 18 kg less water** for non-air-entrained and **15 kg less** for air-entrained concrete. Water-reducing admixtures may cut water by 5% or more.

9.6 Step 4: Water–cement ratio

(a) For strength (ACI Table A1.6.3.4(a)), 28-day cylinder strength:

Strength (MPa)	Non-air-entrained	Air-entrained
40	0.42	–
35	0.47	0.39
30	0.54	0.45
25	0.61	0.52
20	0.69	0.60
15	0.79	0.70

(For ≤ 2% air non-air-entrained and 6% air-entrained; 152 × 305 mm cylinders; aggregate 19–25 mm. For a given w/c, strength rises as maximum aggregate size decreases.)

(b) For durability (Table A1.6.3.4(b)), maximum w/c in severe exposure:

Type of structure	Wet continuously or frequently and exposed to freezing and thawing	Exposed to sea water or sulfates
Thin sections (railings, curbs, sills, ledges, ornamental work) and sections with less than 5 mm cover	0.45	0.40
All other structures	0.50	0.45

(If sulfate-resisting cement, Type II or V, is used, the limit for sulfates may be increased by 0.05.)

With pozzolans (w/(c + p)): two approaches.

1. **Equivalent weight:** w/(c + p) is taken **equal to the w/c** needed. The pozzolan fraction by weight is $F_w = p \div (c + p)$.
2. **Equivalent absolute volume:** keeps the same volume of cementitious material:

$$F_w = \frac{1}{1 + (3.15 \div G_p)(1 \div F_v - 1)} \qquad F_v = \frac{1}{1 + (G_p \div 3.15)(1 \div F_w - 1)}$$

where F_v = pozzolan fraction by absolute volume and G_p = specific gravity of the pozzolan.

Worked example (from the class)

w/c = 0.6, water = 270 kg/m^3, fly ash = 20%.

- **By weight:** c + p = 270 ÷ 0.6 = 450 kg. Pozzolan = 0.20 × 450 = **90 kg**; cement = 450 − 90 = **360 kg**.
- **By absolute volume** (20% by volume, G_p = 2.40): F_w = 1 ÷ [1 + (3.15 ÷ 2.4)(1 ÷ 0.2 − 1)] = 1 ÷ (1 + 1.3125 × 4) = **0.16**. Pozzolan = 0.16 × 450 = **72 kg**; cement = 450 − 72 = **378 kg**.

9.7 Step 5: Cement content

Cement = mixing water (Step 3) ÷ w/c (Step 4). If the specification also gives a **minimum cement content**, use whichever gives **more** cement. Pozzolans and chemical admixtures affect both fresh and hardened properties.

9.8 Step 6: Coarse aggregate content

ACI Table A1.6.3.6, **volume of dry-rodded CA per unit volume of concrete**:

Max. size (mm)	FM 2.40	FM 2.60	FM 2.80	FM 3.00
9.5	0.50	0.48	0.46	0.44
12.5	0.59	0.57	0.55	0.53
19	0.66	**0.64**	0.62	0.60
25	0.71	0.69	0.67	0.65
37.5	0.75	0.73	**0.71**	0.69
50	0.78	0.76	0.74	0.72
75	0.82	0.80	0.78	0.76
150	0.87	0.85	0.83	0.81

Dry mass of CA = volume × dry-rodded bulk density. For less workable concrete (pavements) the volume may be increased by about 10%; for pumping, reduced

by up to 10%.

9.9 Step 7: Fine aggregate content

(a) Mass basis: FA = (estimated mass of fresh concrete) − (water + cement + CA). First estimate of fresh concrete mass (ACI Table A1.6.3.7.1):

Max. size (mm)	Non-air-entrained (kg/m³)	Air-entrained (kg/m³)
9.5	2280	2200
12.5	2310	2230
19	**2345**	2275
25	2380	2290
37.5	**2410**	2350
50	2445	2345
75	2490	2405
150	2530	2435

or calculated: $U_M = 10\,G_a(100 - A) + C_M(1 - G_a \div G_c) - W_M(G_a - 1)$, where G_a = average SG of combined aggregate (SSD), G_c = SG of cement (3.15), A = air %, W_M = water, C_M = cement (kg/m³).

(b) Absolute volume basis (more accurate): FA volume = 1 − (volumes of water + cement + CA + air); FA mass = volume × SG × 1000.

9.10 Step 8: Moisture adjustment

Same idea as the IS method: wet aggregate masses = dry × (1 + total moisture); added water = design water − **free (surface) water** on the aggregates, where surface water = total moisture − absorption.

9.11 ACI 211.1 example (from the slides)

Data: required average strength **24 MPa**; slump 75–100 mm; CA 37.5 mm, dry-rodded mass 1600 kg/m³, SG 2.68, absorption 0.5%; FA SG 2.64, absorption 0.7%, FM 2.8; Type I cement SG 3.15; non-air-entrained (not exposed to severe weathering).

- Step 1: slump 75–100 mm. Step 2: 37.5 mm.
- Step 3: water = **181 kg/m³** (entrapped air 1%).
- Step 4: w/c for 24 MPa = **0.62**.
- Step 5: cement = 181 ÷ 0.62 = **292 kg/m³**.
- Step 6: CA = 0.71 m^3 × 1600 = **1136 kg**.
- Step 7 (mass basis): fresh concrete 2410 kg; FA = 2410 − (181 + 292 + 1136) = 2410 − 1609 = **801 kg**.
- Step 7 (volume basis): water 0.181 + cement 292 ÷ 3150 = 0.093 + CA 1136 ÷ 2680 = 0.424 + air 0.010 = 0.708 m^3; FA = 1 − 0.708 = 0.292 m^3 → 0.292 × 2.64 × 1000 = **771 kg**.

Per m^3	Mass basis	Absolute volume basis
Water (net mixing)	181	181
Cement	292	292
Coarse aggregate (dry)	1136	1136
Sand (dry)	801	771

- Step 8: moisture 2% in CA and 6% in FA (mass basis): CA (wet) = 1136 × 1.02 = **1159 kg**; FA (wet) = 801 × 1.06 = **849 kg**. Surface water: CA 2 − 0.5 = 1.5%; FA 6 − 0.7 = 5.3%. Water to add = 181 − 1136 × 0.015 − 801 × 0.053 = **122 kg**.
- **Batch masses:** water 122, cement 292, CA (wet) 1159, FA (wet) 849, **total 2422 kg/m³**.

9.12 Example 1 (practice problem, solved)

Given: below-ground structure, not exposed to severe weathering or sulfates (so **non-air-entrained**); average 28-day strength **25 MPa**; CA max size **40 mm** (use the 37.5 mm column), dry-rodded density 1600 kg/m^3, SG (SSD) 2.68, absorption 0.5%, total moisture 2%; FA SG 2.65, absorption 0.7%, total moisture 6%, FM 2.80.

1. **Slump:** foundation walls/footings → max 75 mm; adopt **75 mm** (75–100 row).
2. **Max size:** 37.5 mm.
3. **Water: 181 kg/m³**; entrapped air **1%**.

4. **w/c:** 25 MPa → **0.61**.
5. **Cement:** 181 ÷ 0.61 = **297 kg/m³**.
6. **CA:** FM 2.80, 37.5 mm → 0.71 × 1600 = **1136 kg**.
7. **FA (absolute volume):** water 0.181 + cement 297 ÷ 3150 = 0.094 + CA 1136 ÷ 2680 = 0.424 + air 0.010 = 0.709 m^3 → FA = 0.291 × 2.65 × 1000 = **771 kg**. (Mass basis would give 2410 − 1614 = 796 kg.)
8. **Moisture:** CA (wet) = 1136 × 1.02 = **1159 kg**; FA (wet) = 771 × 1.06 = **817 kg**; water = 181 − 1136 × 0.015 − 771 × 0.053 = 181 − 17.0 − 40.9 = **123 kg**.

Batch per m³: water 123 kg, cement 297 kg, CA (wet) 1159 kg, FA (wet) 817 kg.

9.13 Example 2 (practice problem, solved)

Given: exterior column; f'_c = **20 MPa**, not more than **1 in 20** results below it; slump **75 mm**; max size **20 mm** (use 19 mm); FM 2.60; SG of both aggregates 2.65; negligible absorption and moisture; CA bulk density 1600 kg/m^3. Assume **non-air-entrained** (no freezing mentioned).

1. **Required average strength:** "1 in 20" means k = 1.64, so f'_{cr} = 20 + 1.64 s. No standard deviation is given; with no data ACI 318 uses f'_c + 7 = **27 MPa** (the same as assuming s ≈ 4 MPa: 20 + 1.64 × 4 = 26.6).
2. **Water:** 19 mm, 75–100 mm slump → **205 kg/m³**; entrapped air **2%**.
3. **w/c:** interpolate between 30 MPa (0.54) and 25 MPa (0.61): 0.61 − (2 ÷ 5) × 0.07 = **0.58**.
4. **Cement:** 205 ÷ 0.58 = **353 kg/m³**.
5. **CA:** 19 mm, FM 2.60 → 0.64 × 1600 = **1024 kg**.
6. **FA (absolute volume):** water 0.205 + cement 353 ÷ 3150 = 0.112 + CA 1024 ÷ 2650 = 0.386 + air 0.020 = 0.724 m^3 → FA = 0.276 × 2.65 × 1000 = **732 kg**. (Mass basis: 2345 − 1582 = 763 kg.)
7. No moisture correction is needed.

Batch per m³: water 205 kg, cement 353 kg, FA 732 kg, CA 1024 kg (≈ 1 : 2.07 : 2.90, w/c 0.58).

Quick revision

- Good concrete: strength, durability, workable, economical. Fresh: uniform, fills forms, compacts easily, no segregation, finishes well.
- IS 456 Table 5 (RCC): mild 300/0.55/M20; moderate 300/0.50/M25; severe 320/0.45/M30; very severe 340/0.45/M35; extreme 360/0.40/M40. Max OPC 450 kg/m^3.
- f'_{ck} = higher of f_{ck} + 1.65 S and f_{ck} + X. S: 3.5 / 4.0 / 5.0 / 6.0; X: 5.0 / 5.5 / 6.5 / 8.0. Fair control: S + 1.
- IS steps: target → w/c (Fig. 1 curves, check IS 456) → air (1% for 20 mm) → water (186 kg for 20 mm, 50 mm slump; +3% per 25 mm; −10/15/20 kg for shape; −5–10% WRA; −20–30% SP) → cement = W ÷ (w/c) → CA fraction (0.62 for 20 mm Zone II at 0.5; ±0.01 per 0.05) → volumes → masses → moisture correction → trials.
- HSC: S = 6, X = 8; air 0.5% (20 mm); w/cm from Table 8; SF 5–10%, FA 15–30%, GGBS 25–50%, MK 5–15%; CA fraction 0.66 (20 mm Zone II at 0.30).
- ACI: slump → max size → water and air → w/c (strength and durability) → cement → CA (dry-rodded volume × density) → FA (mass or absolute volume) → moisture.
- ACI f'_{cr}: larger of f'_c + 1.34 s and f'_c + 2.33 s − 3.5 (≤ 35 MPa).

Questions to practise

1. What are the requirements of a good concrete mix in the fresh and hardened states? (5 marks)
2. Explain characteristic strength and target mean strength with a sketch of the normal distribution. (5 marks)
3. List the data required for mix proportioning as per IS 10262:2019. (4 marks)
4. Describe the step-by-step procedure of mix design as per IS 10262:2019. (10 marks)
5. Design an M30 mix: OPC 53, 20 mm crushed angular aggregate, Zone II sand, moderate exposure, slump 100 mm, good control, superplasticizer giving 25%

water reduction. SG: cement 3.15, CA 2.74, FA 2.65; aggregates in SSD condition. (15 marks)

6. How does mix design for high-strength concrete differ from that for normal concrete? (6 marks)
7. Describe the ACI 211.1 method of mix design. (10 marks)
8. Explain the equivalent-weight and equivalent-absolute-volume approaches for pozzolans in ACI mix design, with an example. (6 marks)
9. Using the ACI method, design a mix for an average strength of 30 MPa, slump 50 mm, 19 mm aggregate (dry-rodded density 1580 kg/m^3, SG 2.70), FA FM 2.60 and SG 2.65, non-air-entrained. (12 marks)